Market Information Processing in New Product Development: The Importance of Process Interdependency and Data Quality

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***Abstract*—While there is a growing interest in the role of market information processing activities (i.e., the acquisition, dissemination, and use of market information) in new product development (NPD), a number of gaps remain in our knowledge on this topic. When investigating the performance impact of market information processing, most studies have treated the three activities as independent. Our research adds to the extant knowledge by exploring not only both direct relationships between each of the market information processing activities and new product performance, but also interaction effects. We, thus, ask the question of whether there may be synergies in obtaining performance increases by jointly improving two processing activities, rather than just considering each activity independently. In addition, we investigate these effects for different levels of information quality; a topic largely neglected in the market information processing literature. Our analysis is based on empirical data from 152 Dutch NPD projects. The results indicate that market information acquisition and use are both directly associated with increased performance. We also find significant interaction effects for information acquisition and dissemination, and for information dissemination and use. Finally, the importance of information quality is emphasized, with lower quality information producing lower performance and wiping out the effects between various aspects of market information processing and new product performance. We provide several implications of our findings for managers and academics.**

***Index Terms*—Data quality, information acquisition, information dissemination, information use, market information processing, new product performance.**

I. Introduction

THE CONCEPT of a market orientation has been a heavyweight on the research agenda in marketing and new product development (NPD) for over three decades. It has been conceptualized from both behavioral and cultural perspectives [36]. The cultural market orientation perspective focuses on organizational norms and values that encourage behaviors that are consistent with a focus on customers and competitors [16], [54]. The behavioral perspective concentrates on organizational activities that are related to the generation and dissemination of, and responsiveness to, market information (e.g., [42]).

The theoretical foundation of the behavioral perspective on market orientation is the information-processing approach in organizational theory (e.g., [19], [22], [23], [62], [64]). Effective information processing is viewed as consisting of three activities: knowledge acquisition, knowledge dissemination, and knowledge use (e.g., [9], [13], [65]). A basic assumption of this information-processing approach is that organizations are open systems that must deal with uncertainties originating from a changing environment. Efficient information-processing mechanisms can help firms coping with uncertainty by keeping track of changes in the marketplace, alerting the organization about these changes, evaluating their potential effects and acting in anticipation of or reacting to these changes [62].

There is a vast literature on the importance of market orientation to firm performance, both from a cultural as well as from a behavioral perspective (e.g., [37], [42], [47], [54]). In addition, there is an increasing literature on the positive effect of a market orientation on new product performance [6], [25], [44]. However, our understanding of how the different market information processing activities (i.e., acquisition, dissemination and use) influence new product performance is limited [52], [56], [63]. This is surprising as the amount and quality of the information that is processed by NPD managers determines the outcomes and quality of a wide array of decisions for the NPD project, and thus the product's subsequent market performance.

In addition, it is unclear if the different market information processing activities interact synergistically in generating new product performance [38], [41]. This is unfortunate, as we do not understand sufficiently yet if and how interdependencies between the market information processing activities complement each other to produce successful new product outcomes. Finally, the extant literature is silent on the potential role of the quality of the market information that is processed, and how this data quality possibly moderates the impact of different market information processing activities on new product performance.

This article addresses these issues and aims to contribute to the extant literature by developing and testing a model that investigates the main and interaction effects of the three market information processing activities on new product performance. We investigate whether and under which circumstances an increased level of proficiency in market information processing activities manifests in increased performance. In our analyses,

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we control for varying degrees of product newness and market predictability, as these constructs could be responsible for performance differences as well.

In addition to the direct performance effects of the market information processing activities on performance, we also analyze interdependencies between information acquisition, dissemination and use. Potential interdependencies between these activities have not yet been empirically explored. Until now, the constructs have been treated as distinct and independent subdimensions of market information processing, while their combined effects have been neglected. From an NPD manager's viewpoint this investigation is relevant as not only does management need evidence on how effective information use in each stage of the NPD process is for performance (see, for instance, [63]), but they also need information on the interactions between different information processing activities and performance. In management practice, the different activities that constitute information processing may not be independent. For example, information cannot be distributed unless it has first been acquired. This means that evidence on the question of whether these different information-processing activities reinforce or weaken each other may be valuable in making, for instance, resource allocation decisions.

We also test how these relationships change based on information quality (i.e., its accuracy, objectiveness, clarity, and timeliness). To achieve this, we divide our sample into two groups: one group where the data that were processed were of below average quality in the sample, and the second group where the data quality was above the sample's average. Hence, we provide evidence as to whether information-processing activities have the same performance effects for different levels of data quality.

The rest of this paper is organized as follows. The following section develops our conceptual model and discusses the proposed relationships. Subsequently, we describe the methodology used in the empirical study. Then, we present the results. The article concludes with a discussion of the main findings, limitations of the study, and future research directions.

II. Developing the Conceptual Model

Two recent meta-analyses show that market information processing in aggregate has a positive impact on firm performance and new product performance [31], [41]. Market information processing enhances a firm's NPD-related activities and performance because it drives a proactive disposition toward meeting customer needs and competitors' actions [5], [15], [33], [44]. In addition, market information processing should enhance new product performance because it involves doing something new or different in response to market conditions [39]. Market-oriented firms emphasize learning to uncover latent customer needs, and that learning enhances their ability to create and implement new ideas, products, and processes [35], [55].

From the literature review, two points seem to merit specific attention. Although some studies investigate market information processing in the innovation context, the focus mostly is on the firm or program level. In addition, few studies have investigated the impact of distinct market information processing activities on performance indicators. Hence, there is a lack of evidence for the relationship between individual information processing activities and the performance of a specific product—which is expected to be highly valuable for NPD managers.

Previous research has shown that there is a temporal flow to market information processing: information must be acquired before it can be disseminated or used [63]. When individuals on the NPD team acquire market information themselves, this enhances each individual's understanding of the issues relevant for the specific product [29]. McQuarrie [49] found that when individuals on a development team directly acquired information about customer needs, they ascribed a higher credibility to it. Furthermore, as Leonard and Rayport [46] suggest, although developers may not consciously realize it, they may empathically use information they have acquired from customers in making relevant decisions when they develop new products. From this reasoning, we propose:

H1: Information acquisition is positively related to new product performance.

Frequently in firms, acquiring market information is predominantly the responsibility of the marketing function. However, others on the NPD team, especially the engineers, technology experts, and controllers, also need this information in order to create a product that has the right features and price. Im and Workman [37] suggest that dissemination has a positive effect on new product performance as it creates a common understanding of market developments across NPD team members from all functions. Additionally, Gatignon and Xuereb [25] proposed that sharing market information is crucial for NPD, as it helps the team become aware of potential problems and enhances problem solving. Finally, Moorman and Miner [53] empirically found that high levels of knowledge transfer among team members are positively associated with effective and efficient decision-making, and in turn, NPD financial performance. However, Henard and Szymanski [34] found that excessive collaboration and information sharing can produce a negative performance effect. Overall, we find the reasoning of Im and Workman [37] and Gatignon and Xuereb [25] compelling, and thus hypothesize:

H2: Information dissemination is positively related to new product performance.

Finally, a number of studies previously have found positive empirical relationships between information use and NPD performance [52], [56], [63]. Organizational theory suggests that information use is related to organizational performance because it reduces uncertainty [62]. As NPD is fundamentally about reducing technical and marketing uncertainties as the project progresses through the development process, we propose:

H3: Information use is positively related to new product performance.

Despite the available research on the effect of market information processing on new product performance, our understanding of how the different activities of market information processing interact with one another in generating performance is limited. In order to narrow this gap, recent publications call for studies to examine potential interaction effects among market information processing activities. For example, Jimenez-Jimenez and

Cegarra-Navarro [40] note that "the relationship is more complex than a linear one," thus reiterating the need to examine interactions among the different market information processing activities.

Until now, only one empirical study has investigated how the different components of a market orientation complement one another [38]. However, the authors build their analysis on the market orientation concept of Narver and Slater [54], thus using the dimensions of customer orientation, competitor orientation and interfunctional coordination, rather than the different activities of market information processing. Furthermore, they do not investigate interaction effects on NPD performance but on marketing program creativity.

In thinking about potential interdependencies, it is likely that sharing information acquired from the market is crucial for NPD as it establishes a common understanding [25]. One may argue that market information acquisition in and of itself may not be a sufficient condition for new product performance, especially when one function (such as marketing) is predominantly responsible for this task. New data may not always be fully understandable by a single person with their unique mindset, background, and experience. For an adequate assessment of different developments and disparate pieces of information that may impact the team's decisions, different functional experts need to consider different perspectives [20]. Until information acquired from the market is disseminated across the team, it is unlikely to be infused with organizational meaning. Thus:

H4: The interaction of acquisition and dissemination is positively related to new product performance.

The few empirical studies distinguishing between information acquisition, dissemination, and use, and their relationship to NPD performance give further insight into additional potential interdependencies [52], [56], [63]. In all three studies, the acquisition, dissemination and use of market information were positively correlated. While Veldhuizen *et al.* [63] found that both information acquisition and use were positively related to new product performance, Moorman [52] and Ottum and Moore [56] found a significant performance effect of information use only. However, the latter study shows that information use was most strongly related to information sharing, which in turn was related to the amount of information acquired [56]. These findings highlight that information dissemination is only indirectly associated with performance, through its association with information use. The studies of Moorman [52], Ottum and Moore [56], and Veldhuizen *et al.* [63] also offer no theoretical reason why information dissemination in and of itself should lead to more successful products, if the NPD team does not act upon (use) the information that has been disseminated. We thus hypothesize:

H5: The interaction of dissemination and use is positively related to new product performance.

In the most recent study of the differential effects of information processing activities on new product performance, Veldhuizen *et al.* [63] also found that information acquisition, dissemination and use were correlated, and that various aspects of information processing acted differentially in relationship to obtaining increased product performance. Most interestingly, the results of their path analysis suggested that information acquisition may be associated directly with use, rather than needing to be disseminated first before it could be used. As suggested by McQuarrie [49], this may occur when the people who need information to make NPD decisions are in fact the ones who acquire it, eliminating the need for it to be disseminated before it can be used. We thus hypothesize:

H6: The interaction of information acquisition and use is positively related to new product performance.

Although we posit that the components of market information processing are positively related to performance, inaccurate or biased information can produce bad decisions, and unclear information can increase uncertainty rather than reduce it. NPD is essentially an information processing activity [45], and as the old computer saying goes, "garbage in, garbage out." Information can be acquired in inaccurate or biased forms, or it can be biased or rendered inaccurate in its dissemination or use. Additionally, information that has been acquired or disseminated after a decision has been made cannot be used. Thus, information quality matters when making decisions. The current understanding of the role played by information quality when assessing the effectiveness of market information processing is limited. Zahay and Griffin [66] find that having quality information is a key component to having effective customer information systems and in turn key to effective information processing in NPD. Another paper based on this same survey also found a statistically significant relationship between information quality and firm performance [68]. These related empirical findings suggest a moderating effect of information quality on the relationship between information processing activities and new product performance. We thus hypothesize:

H7: Information quality moderates the relationship between information processing activities and new product performance. These relationships will be diminished for projects with lower quality information and attenuated for projects with higher quality information.

III. METHOD

A. Measure Development

We used established constructs and measures from prior studies whenever possible. We measured market information acquisition and dissemination with items adapted from Jaworski and Kohli [39] to reflect the project-level unit of analysis. To measure market information use during the NPD process, a new scale was developed to better represent activities related to the specific new product. The items are based on scales by Jaworski and Kohli [39], Menon and Varadarajan [50], and Ruekert [58]. New product performance was measured with items adapted from Griffin and Page [30]. The moderating construct of market information quality was measured with items provided by Moenaert *et al.* [51]. We used three constructs to control for the newness of the products in our sample: technological newness of the product, market newness of the product, and newness of the product to the company. These constructs were adapted from Avlonitis *et al.* [7], Danneels and Kleinschmidt [14], Gatignon

TABLE I
SAMPLE CHARACTERISTICS ($N = 152$)

Industry		**No. of employees**		**Sales (in EUR Mio.)**	
Chemical & Pharmaceutical	15.3 %	<51	16.7 %	<2	6.7 %
Electrical & Industrial Machinery	47.8 %	51-75	19.4 %	2-6	14.4 %
Computers & Electronics	12.6 %	76-100	12.0 %	6-10	16.7 %
Medical Appliances & Optical Instruments	10.8 %	101-150	23.2 %	10-15	14.4 %
Information Technology	13.5 %	151-300	20.4 %	15-25	13.4 %
		>301	8.3 %	25-50	14.4 %
				50-100	15.6 %
				>100	4.4 %

et al. [24], and Green *et al.* [27]. We measured market predictability with items proposed by Atuahene-Gima [6].

To test the initial properties of the measurement scales, the questionnaire was subjected to a quantitative pilot survey. We contacted 110 NPD managers of high-tech firms from a commercial mailing list by phone and 57 agreed to participate. We sent them the questionnaire including all the constructs. They were asked for comments on the questions and to rate the topic's overall relevance. The results showed that managers considered the questions to be clear and the topic to be relevant. Based on the results of the pilot survey, we made minor changes in wording in order to increase the use of upper and lower bounds of our scales.

B. Sample and Data Collection

We gathered our data in The Netherlands using a cross sectional survey methodology. We used the project level of analysis because the information used to make NPD decisions is project-specific, rather than being more generally applicable to the entire firm or a division. Each project requires specific details about customer needs and technical capabilities to allow developers to make tradeoffs between the different specifications and feature levels.

The research encompassed the following high-tech industries: chemical and pharmaceutical, electrical and industrial machinery, computers and electronics, medical appliances, optical instruments, and information technology. We targeted high-tech industries as firms in these industries engage in intense NPD activities, they tend to be more complex products, requiring more information than simpler products, and information about technologies, customer needs and competitive offerings changes more quickly than in lower technology industries.

We fielded the research using a sample drawn from the Review and Analysis of Companies in Holland (REACH) directory. Our target respondents were NPD managers. We asked these managers to report on a project in which they had been involved as project leader or as the main marketing expert. The product was launched into the market no longer than three years ago to reduce retrospective bias. Five hundred fifty potential respondents were prenotified by phone in order to explain the intention of the study, encourage commitment, and secure the respondents' familiarity with the topic. All respondents received a questionnaire via mail. Up to three follow-up contacts to nonrespondents were made by phone and postcards. In total, 152 completed questionnaires were returned for an effective response rate of 27.6%. Table I shows the sample characteristics.

To detect potential problems of nonresponse bias, the dataset was divided into thirds, according to the number of days from initial mailing until receipt of the questionnaire [2]. As the results of the t-test between the first and the last thirds indicated no statistically significant differences (at the $p < 0.05$ level) in the mean responses for any of the constructs, we concluded that nonresponse bias was not a major reason for concern. In addition, we compared the companies in our sample with those firms that did not respond on several demographic variables, such as firm size and industry type. No significant differences were found. Potential concerns about common method bias were alleviated by conducting Harman's one-factor test in which all variable items were entered into a single principal component analysis (PCA) [57]. The results showed that neither a single factor nor one general factor accounted for the majority of the covariance in the items.

C. Measure Validation

The operational definitions of the constructs in this paper suggest formative rather than reflective measurement since the indicators are viewed as defining characteristics of the constructs [17]. We conceived of all information processing constructs as combinations of various information aspects in each processing stage [67]. The acquisition of market information, for instance, describes the extent to which different types of activities were pursued, aiming at getting in-depth market insight. While NPD projects that pursue all activities put a stronger emphasis on acquiring market information than those less active, these activities do not necessarily correlate highly with one another; e.g., a firm may constantly assess industry shifts, but rarely meet with potential customers.[1]

[1] Although most of the existing scales that we adapted in this paper treated their constructs as reflective measures, we believe that from a content perspective, formative measurement is a more appropriate way to treat our constructs. In other words, taking action along each activity in the construct was necessary to obtain the full set of information required for NPD [68]. However, we also tested the constructs as reflective measures. All constructs were unidimensional (CFA), and all were reliable (lowest Cronbach's alpha = 0.67). Most interesting, our results were not influenced by the choice between a formative or reflective construction of our scales.

Respondents in our pretest judged our measures as being thorough in covering the construct domain as well as being understandable, indicating reliable and valid measurement. We used variance inflation factors (VIF) and condition indices (CI) to assess indicator collinearity [8], [18]. For this evaluation, we conducted multiple linear regression analyses for each multi-item construct. The VIF values were well below the threshold of 10 for critical multicollinearity [48]. In addition, the CI extracted, which reflects the relative amount of variance associated with each eigenvalue, remains below the threshold value of 30 [8]. Table II reports all items, VIF, and CI.

In the next step, we computed mean summated scores from the relevant indicators to derive aggregate scales for all constructs. Table III presents descriptive statistics and correlations. The means and standard deviations indicate that there is sufficient variability in our measures to test our hypotheses [32].

IV. Analysis and Results

To eliminate concerns regarding a potential lack of independence among the market information processing constructs, we conducted tests for multicollinearity using the same criteria as described above for the indicators [32]. The results show that the CI and VIF never exceed the critical values of 30 and 10, respectively. Thus, all three constructs can be used for testing their relationship with new product performance. Table IV presents these results.

We included both product- and market-specific control variables in our model. As several studies provide evidence that these variables may be related to new product performance, we considered technological newness of the product, market newness of the product, product newness for the company, and market predictability as controls (e.g., [14], [61]). Only market predictability was found to significantly drive performance ($\beta = 0.187$, $p < 0.05$), indicating that new products being developed for foreseeable environments are more likely to be successful than those entering less predictable markets.

Information acquisition and information use exhibit significant and positive regression coefficients ($\beta = 0.215$, $p < 0.05$ and $\beta = 282$, $p < 0.01$, respectively), supporting H1 and H3. However, dissemination of market information was not found to have a significant impact on new product performance ($p > 0.05$), producing no support for H2.

The next step in the analysis tested for interaction effects between the market information processing activities, which would indicate their complementary nature. We used hierarchical regression analysis [1] for these tests. Table V displays the results. After regressing the four control variables and the three market information processing activities on new product performance in regression model 1, we introduced the three interaction terms in models 2–4. Prior to creating the interaction terms, the variables were mean-centered to reduce potential problems of multicollinearity.

Comparing regression models 1 through 4, we find that when the interaction variables for models 2 and 3 enter the regression equation, the adjusted R^2 increases significantly. The coefficients for information acquisition and dissemination and information dissemination and use are statistically significant ($\beta = 0.172$, $p < 0.05$ and $\beta = 0.202$, $p < 0.05$, respectively), supporting H4 and H5. The information acquisition and use interaction effect is not statistically significant ($p > 0.05$). We also find that the information acquisition and use term does not provide additional explanatory power, as indicated by the decreased F-value. This result implies that the effect between information acquisition and use is an artifact, inhibiting further interpretation [12].

As a last step in the analysis, we split the sample into two groups based on market information quality. The median of information quality (IQ = 3.00) was used for determining group affiliation. Respondents for projects with an IQ < 3.00 ($n = 67$) disagreed with statements that the quality of their information was objective, accurate, clear to everyone, and timely at hand. The number of respondents that agree with those statements (IQ $\geq$ 3.00) was 82 cases. To assess whether the market information processing activities and their interaction effects were of different relevance for the two different groups, the hierarchical regression analyses investigating each interaction effect were rerun for both subsamples. Table VI shows the results.

We find that the results differ across the two subgroups, both in terms of the direct relationships between aspects of information processing and performance, and then also with the magnitude of the interaction effects, as hypothesized. For the low information quality group, all previously found associations are eliminated. No aspect of information processing is associated with performance. For NPD teams with high quality information available during the project, the direct effects of information acquisition and use, as well as the interaction effect between information dissemination and use, manifest again. As predicted, the effect sizes, levels of statistical significance and variance accounted for by the variables are higher for each of the high-quality equations than they are for the low-quality group. H7, where we had predicted that data quality would moderate the relationships between the information processing activities and performance, is thus fully supported by our results.

V. Discussion and Managerial Implications

Previous empirical research investigating the relationships between market information processing activities and new product performance has produced rather one-dimensional results. For example, both Moorman [52] and Ottum and Moore [56] found that information acquisition, dissemination, and use were correlated, but that only information use was associated with performance. The results presented here go a bit further. As with these two previous investigations, we also found that these three market information processing activities were correlated. In the full sample, however, both information acquisition and use are associated with performance. We also find that they act differentially on performance, depending upon the quality of the information available to the team. When the sample is split into two subgroups based on information quality, information acquisition and use are associated with performance only for teams with higher information quality. The relationships between information processing and performance become insignificant in the

TABLE II
MEASUREMENT MODELS

Constructs	VIF	CI	Source
Acquisition of Market Information: During the NPD project…			
We regularly met with potential customers to find out their future needs.	1.64	12.89	Jaworski and Kohli (1993)
We polled end users several times to assess the quality of our product.	1.72		
Intelligence on our competitors was generated by different departments.	1.51		
We were quick in detecting fundamental shifts in our industry.	1.50		
We periodically reviewed the likely effect of changes in our business environment on customers.	1.65		
Dissemination of Market Information: During the NPD project…			
A lot of informal 'hall talk' concerned our competitors' tactics or strategies.	1.23	13.29	Jaworski and Kohli (1993)
We had interdepartmental meetings to discuss market trends and developments.	1.60		
Data on customer satisfaction were disseminated at all levels on a regular basis.	1.61		
Employees spent time discussing customers' future needs.	1.21		
There was a lot of communication between project members concerning market developments.	1.47		
Use of Market Information: During the NPD project…			
Market information was used in solving project-related problems.	1.14	12.24	Jaworski and Kohli (1993); Menon and Varadarajan (1992); Ruekert (1992)
Market information was taken into account in decisions about the new product.	1.67		
Market information was used to improve the new product.	1.53		
Market information was used to tailor the new product to market needs.	1.21		
NPD Performance: The new product attained…			
Unit sales goals.	2.69	17.31	Griffin and Page (1993)
Revenue growth goals.	2.90		
Market share goals.	2.25		
Sufficient sales as a percentage of total company sales.	2.83		
Market Information Quality: Market information that was available during the NPD project…			
Was objective.	1.68	15.37	Moenaert et al. (1994)
Was accurate.	1.98		
Was clear to everyone.	2.76		
Came exactly at the right time.	2.59		
Technical Newness of Product: The new product…			
Contains a lot of technical knowledge.	1.52	17.84	Danneels and Kleinschmidt (2001), Green et al. (1995)
Is technically complex.	1.70		
Contains sophisticated technology.	1.91		
Has a technologically advanced character.	2.28		

TABLE II
(CONTINUED)

Market Newness of Product: When the new product was launched to the market…			
It had features that did not exist yet.	1.67	17.68	Danneels and Kleinschmidt (2001), Gatignon et al. (2002) and Green et al. (1995)
There were no comparable products on the market.	1.78		
It was new for the product category.	1.85		
It was the first of its kind.	2.51		
It was innovative to the market.	2.06		
Newness of Product to the Company			
The customers for this product were new to our company.	1.52	10.93	Avlonitis et al. (2001), Danneels and Kleinschmidt (2001), Gatignon et al. (2002)
The competitors we face in the market were new to our company.	2.80		
The product category itself was new to our company.	1.91		
The marketing strategy for the product was new to our company.	1.68		
The competitive environment for the product was new to our company.	2.69		
Market Predictability			
Developments in the market for the new product were predictable.	1.65	11.23	Atuahene-Gima (1995)
Changes in product preferences of customers were predictable.	1.54		
Future needs of customers were predictable.	1.33		

TABLE III
DESCRIPTIVE STATISTICS AND CORRELATIONS OF CONSTRUCTS ($N = 152$)

				Correlations								
	Constructs	**Mean**	**S.D.**	1	2	3	4	5	6	7	8	9
1	*Acquisition of Market Information*	3.05	.76	1.00								
2	*Dissemination of Market Information*	2.76	.68	**.53****	1.00							
3	*Use of Market Information*	3.24	.78	**.52****	**.61****	1.00						
4	*NPD Performance*	3.28	.80	**.35****	**.25****	**.39****	1.00					
5	*Market Information Quality*	2.87	.73	**.43****	**.41****	**.46****	**.26****	1.00				
6	*Technical Newness of Product*	3.61	.74	**.39****	**.20***	**.18***	.08	.14	1.00			
7	*Market Newness of Product*	3.53	.89	**.16***	.02	.01	.07	.11	**.20***	1.00		
8	*Newness of Product to the Company*	2.39	.95	.07	**.17***	.13	-.04	.01	.09	-.03	1.00	
9	*Market Predictability*	3.18	.77	.13	-.01	.12	**.25****	**.16***	.04	-.02	-.08	1.00

*p <0.05. ** p <0.01. Significant entries are presented in bold.
1 = Strongly disagree, 5 = strongly agree.

subsample with low information quality. In addition, the interaction effect between information dissemination and use is only significantly related to performance for teams with high-quality information.

The managerial imperative, then, is for NPD teams and managers to determine the quality level of the information that is coming into the team. For example, rather than just accepting a market research report at face value, they should perhaps do some due diligence into the design and execution of the research to ensure that the appropriate people were interviewed or surveyed, in an appropriate manner. Rather than blithely accepting a sales person's insistence that customers need some particular feature, perhaps they should also talk to potential customers themselves to validate the information. And when

TABLE IV
RELATIONSHIP BETWEEN INFORMATION PROCESSING AND NEW PRODUCT PERFORMANCE

	Stand. Regression Coefficients	**VIF**	**CI**
Control Variables			
Technical Newness of Product	-.032		
Market Newness of Product	.047		
Newness of Product to the Company	-.056		
Market Predictability	**.187***		
Information Processing Constructs			
Acquisition of Market Information	**.215***	1.514	
Dissemination of Market Information	-.018	1.785	13.401
Use of Market Information	**.282****	1.723	
R^2	.234		
Adjusted R^2	.196		
F	6.226**		

*$p < 0.05$. ** $p < 0.01$. Significant entries are presented in bold.

reviewing secondary data from the internet or a popular press source, perhaps the team should dig into the information to ascertain the actual source of the information, and its validity. When a team finds that the quality of the data they have gathered or that has been given to them is not high, they need to disregard that information, and seek higher quality data.

Our results thus imply: Market information quality is crucial! Possessing and using higher quality information is associated with higher new product performance (with a mean value of 3.45 versus 3.08 for those with lower quality information; $p < 0.01$). While these results might seem to be utterly predictable at first, in reality few academics have taken information quality into account when investigating market information processing; Zahay and Griffin [66] and Zahay *et al.* [68] being an exception. Additionally, in practice we see few NPD managers who focus on ensuring that the information their teams receive is of the highest possible quality. Based on these results, however, obtaining high information quality would seem to be a first priority before spending valuable resources on any further processing activity. NPD teams need to know how to assess the quality of information that is presented to them.

NPD teams also may need to learn the art and science of doing reliable marketing research and competitive analysis. Information acquisition in and of itself is directly associated with performance only when information quality is high. Thus, NPD teams may benefit by direct contact with potential end users and customers and by being personally involved with understanding competitive and environmental changes—when the people on the team have the technical skills required to gather high-quality information. Therefore, NPD managers first must ensure that teams know how to effectively acquire market information, and second, during development, managers should encourage these teams to uncover information themselves, and provide them with the financial and time resources to do so. This also means that customer and market information acquisition should not be seen as being within the sole purview of the NPD team's marketing experts. Firms may benefit by teaching information gathering and processing capabilities across the NPD team members.

If both acquisition and use are associated with performance for high quality information, then why is there no direct performance effect for disseminating information? One reason may be that only creating a common understanding or generating an awareness of project-specific problems has no impact on performance by itself. The "right" information has to be acquired to make its diffusion a relevant activity. Or, the diffusion has to be followed by taking adequate action. This rationale is supported when turning to the results pertaining to the interaction effects. The lack of significance for the main effect does not mean that dissemination is unimportant, as both the interaction effects with acquisition and use produce significant positive relationships with performance, with more significant effects only found for teams using higher quality information. These results provide evidence that, rather than having a direct effect on NPD performance, dissemination works synergistically through its interactions with acquiring and using information to improve performance.

To elaborate further on these findings, when information is acquired by an individual on an NPD team and then shared among other NPD team members, it may be subjected to being

TABLE V
INTERACTION EFFECT BETWEEN INFORMATION PROCESSING CONSTRUCTS

	Stand. Regression Coefficients			
	Model I	**Model II**	**Model III**	**Model IV**
Control Variables				
Technical Newness of Product	-.032	-.043	-.054	-.042
Market Newness of Product	.047	-.067	-.062	-.052
Newness of Product to the Company	-.056	.056	.037	..045
Market Predictability	**.187***	.139	.124	**.179***
Information Processing Constructs				
Acquisition of Market Information	**.215***	**.233***	**.241***	**.226***
Dissemination of Market Information	-.018	-.021	-.023	-.019
Use of Market Information	**.282****	**.311****	**.337****	**.292****
Interaction Effects				
Acquisition x Dissemination of Market Information		**.172***		
Dissemination x Use of Market Information			**.202***	
Acquisition x Use of Market Information				.056
R^2	.234	.260	.267	.236
Adjusted R^2	.196	.218	.226	.193
ΔR^2		**.026***	**.033***	.002
F	**6.226****	**6.226****	**6.476****	**5.493****

*p <0.05. ** p <0.01. Significant entries are presented in bold.

critically discussed and reflected upon by people with different backgrounds, capabilities and experiences prior to being used, which increases overall understanding of the meaning of the information. Learning theory indicates that raw information that comes into an organization must be interpreted by others in the organization to be truly useful [59]. In the case of information being applied to the project, perhaps a shared understanding needs to be created as to how it applies in this particular project context. Information that has not been shared with others to allow interpretation may have data quality or applicability issues that may not have been detected before that information was applied to the project at hand.

The implications of these two sets of results are twofold. First, just moving information already available in the firm (i.e., there has been no additional information acquisition prior to dissemination), may not be adequate to provide the knowledge that is actually needed by the NPD team to enable them to create successful new products, even if that information is of high quality (objective, accurate, clear, and timely). For example, information that the marketing or market research function has acquired or generated in the past may not be applicable to the current project. Thus, it is only moving information that provides needed knowledge to the NPD project that is important, rather than moving random pieces of information around just to move them. New product performance is increased only when new information that is applicable to this project is acquired and then moved to those who need it or when information (new or old) that is useful to the team is moved to those on the project who need it. Moving old or less useful information around the organization may just be making work rather than improving performance.

Second, as suggested by Song *et al.* [60], this may imply the usefulness of an information "gate-keeping" mechanism, in which someone on the NPD team is formally responsible for assessing that only "the right" (i.e., applicable and, therefore, useful) information gets disseminated to the NPD team, especially when that information is sourced from elsewhere within

TABLE VI
IMPACT OF INFORMATION QUALITY ON THE RELATIONSHIP BETWEEN INFORMATION PROCESSING AND NEW PRODUCT PERFORMANCE

	Low Quality			High Quality		
Control Variables	**Model LI**	**Model LII**	**Model LIII**	**Model HI**	**Model HII**	**Model HIII**
Technical Newness of Product	-.062	-.066	-.067	.023	-.010	.059
Market Newness of Product	.138	.133	.143	-.001	-.058	.025
Newness of Product to the Company	-.215	-.196	-.195	-.007	-.004	-.021
Market Predictability	.132	.171	.171	.137	.092	.142
Information Processing Constructs						
Acquisition of Market Information	.161	.020	.056	**.248***	**.285****	**.248***
Dissemination of Market Information	.169	.093	.101	-.121	-.179	-.071
Use of Market Information	.217	.234	.213	**.403****	**.350****	**.402****
Interaction Effects						
Acquisition x Dissemination of Market Information	.316			.046		
Dissemination x Use of Market Information		.081			**.233***	
Acquisition x Use of Market Information			.115			-.094
R^2	.185	.147	.151	.322	.354	.327
Adjusted R^2	.075	.031	.036	.245	.281	.251
F	1.675	1.268	1.310	**4.210****	**4.861****	**4.310****

*p <0.05. ** p <0.01. Significant entries are presented in bold.

the organization. Managers need to provide teams with the time they need to create a shared understanding of the information they acquire. Managers also may need to colocate their teams during development to encourage sharing information across the team.

The insignificance of the relationship between the information acquisition and use interaction and new product performance is interesting. This result suggests that a stronger information process may be one that acquires and disseminates information to ensure that "the right" information has been acquired, it is of high quality, and that a common understanding of that information's meaning has been created. This should then be followed by a disseminate and use process whereby that information with a shared understanding is distributed back out to the NPD team members who have the most use for it at that point in the NPD process. From a managerial perspective, then, teams need to have time to not just acquire information, but also time and processes that support them in creating a shared understanding of the information acquired [9], [59], [60]. Colocation, while perhaps disruptive to individuals' lives, may be one way to increase information sharing and interpretation.

NPD managers should not myopically focus on improving one perspective of market information processing in isolation in order to generate successful new products. Instead, it is the combination of particular market information processing activities that seems to provide a firm competitive advantage through efficient market information acquisition and dissemination, and dissemination and use. Additionally, NPD teams need time to personally acquire new information and then work to create a shared understanding of that information before using it in the development project. Finally, this research shows that the importance of market information processing is highly dependent on the quality of the processed information. When market information is more objective, accurate, clear to everyone, and timely at hand, the processing of this information becomes a strong performance driver. Thus, the bottom line for managers and NPD team members appears to be that they closely monitor the quality of information that is acquired, gets sent around, discussed and finally is used as a basis for NPD decisions.

A. Limitations and Avenues for Future Research

As with all research, there are limitations that must be considered when interpreting the results. To start with, the sample consists of Dutch firms. This selection may put constraints on the generalizability of the results to other countries, especially those with Eastern rather than Western cultures. In market orientation research, several cross-cultural studies deal with the question of whether differences exist in the performance effect of market information processing in different countries—and the evidence is mixed. Early studies suggested an American bias, because the strongest market orientation effects were typically found in the United States [10], [26], [28]. However, more recent studies have recorded strong positive results in a variety of non-US settings including Germany [36], the Netherlands [44], Spain [43], and elsewhere, disputing the notion that market orientation is a uniquely American concept. Still, future research may consider testing our framework in other national contexts.

In addition, the sample consists of NPD projects from high-tech industries, which may also constrain the generalizability of our results. The focus on such projects was chosen, since they engage in intense NPD activities, and are likelier to have more fluidity in customer needs and technology changes. However, it would be interesting to uncover whether the effects remain stable for other, traditionally somewhat less NPD-dependent sectors, such as services and several consumer goods.

As this paper focuses on the direct and interaction effects of market information processing activities on the performance of a new product, the understanding of such effects on other NPD-related activities and outcomes is limited. Such investigations may be an interesting avenue for further research. For example, the effects of information processing on NPD aspects like the nature of decision making processes or the quality of marketing plans can be clarified.

As suggested by Podsakoff *et al.* [57], several measures were taken to ensure confidentiality and anonymity of the respondents in order to minimize the effects of common method bias. Still, the study data may suffer from a bias caused by collecting both independent and dependent variables from the same respondent. Future research is advised to strive for using dual respondents to avoid such potential biases.

REFERENCES

[1] L. S. Aiken and S. G. West, *Multiple Regression: Testing and Interpreting Interactions*. Newbury Park, CA: Sage, 1991.

[2] J. S. Armstrong and T. S. Overton, "Estimating non-response bias in mail surveys," *J. Mark. Res.*, vol. 14, pp. 396–402, Aug. 1977.

[3] K. Atuahene-Gima, S. F. Slater, and F. M. Olson, "The contingent value of responsive and proactive market orientations for new product program performance," *J. Prod. Innov. Manage.*, vol. 22, pp. 464–482, 2006.

[4] K. Atuahene-Gima and A. Ko, "An empirical investigation of the effect of market orientation and entrepreneurship orientation alignment on product innovation," *Org. Sci.*, vol. 12, no. 1, pp. 54–74, 2001.

[5] K. Atuahene-Gima, "Market orientation and innovation," *J. Bus. Res.*, vol. 35, no. 2, pp. 93–103, 1996.

[6] K. Atuahene-Gima, "An exploratory analysis of the impact of market orientation on new product performance," *J. Prod. Innov. Manage.*, vol. 12, no. 4, pp. 275–293, 1995.

[7] G. J. Avlonitis, P. G. Papastahopoulou, and S. P. Gounaris, "An empirically-based typology of product innovativeness for new financial services: Success and failure scenarios," *J. Prod. Innov. Manage.*, vol. 18, pp. 324–342, 2001.

[8] D. A. Belsley, *Conditioning Diagnostics: Collinearity and Weak Data in Regression*. New York: Wiley, 1991.

[9] H. Bij, M. Song, and M. Weggeman, "An empirical investigation into the antecedents of knowledge dissemination at the strategic business unit level," *J. Prod. Innov. Manage.*, vol. 20, no. 2, pp. 163–179, 2003.

[10] H. N. Chan and P. Ellis, "Market orientation and business performance: some evidence from Hong Kong," *Int. Mark. Rev.*, vol. 15, no. 2, pp. 119–139, 1998.

[11] G. Chow, "Tests of equality between sets of coefficients in two linear regressions," *Econometrica*, vol. 28, no. 3, pp. 591–605, 1960.

[12] J. Cohen, P. Cohen, S. G. West, and L. S. Aiken, *Applied Multiple Regression/Correlation Analysis for the Behavioral Sciences*. Hillsdale, NJ: Lawrence Erlbaum Associates, 2003.

[13] R. Coombs and R. Hull, "Knowledge management practices and path-dependency in innovation," *Res. Policy*, vol. 27, no. 3, pp. 237–253, 1998.

[14] E. Danneels and E. J. Kleinschmidt, "Product innovativeness from the firm's perspective: Its dimensions and their relation with project selection and performance," *J. Prod. Innov. Manage.*, vol. 18, pp. 357–373, 2001.

[15] L. De Luca and K. Atuahene-Gima, "Market knowledge dimensions and cross-functional collaboration: Examining the different routes to product innovation performance," *J. Mark.*, vol. 71, pp. 95–112, Jan. 2007.

[16] R. Deshpandé, J. U. Farley, and F. C. Webster, "Corporate culture, customer orientation, and innovativeness in Japanese firms: A quadrad analysis," *J. Mark.*, vol. 57, pp. 23–27, 1993.

[17] A. Diamantopoulos and J. A. Siguaw, "Formative versus reflective indicators in organizational measure development: A comparison and empirical illustration," *Brit. J. Manage.*, vol. 17, pp. 263–282, 2006.

[18] A. Diamantopoulos and H. M. Winklhofer, "Index construction with formative indicators: An alternative to scale development," *J. Mark. Res.*, vol. 38, no. 2, pp. 269–277, 2001.

[19] W. R. Dill, "The impact of environment on organizational development," in *Readings in Organization Theory: Open-System Approaches*, J. G. Maurer, Ed. New York: Random House, 1971, pp. 81–93.

[20] D. Dougherty, "Interpretive barriers to successful product innovation in large firms," *Org. Sci.*, vol. 3, no. 2, pp. 179–202, 1992.

[21] P. Ellis, "Market orientation and performance: A meta-analysis and cross-national comparisons," *J. Manage. Stud.*, vol. 43, no. 5, pp. 1089–1107, 2006.

[22] J. R. Galbraith, *Designing Complex Organizations*. Reading, MA: Addison-Wesley, 1973.

[23] J. R. Galbraith, "Designing the innovating organization," *Org. Dyn.*, vol. 10, pp. 5–25, 1982.

[24] H. Gatignon, M. L. Tushman, W. Smith, and P. Anderson, "A structural approach to assessing innovation: Construct development of innovation locus, type, and characteristics," *Manage. Sci.*, vol. 48, no. 9, pp. 1103–1122, 2002.

[25] H. Gatignon and J.-M. Xuereb, "Strategic orientation of the firm and new product performance," *J. Mark. Res.*, vol. 34, no. 1, pp. 77–90, 1997.

[26] B. Gray, S. Matear, C. Boshoff, and P. Matheson, "Developing a better measure of market orientation," *Eur. J. Mark.*, vol. 32, no. 9, pp. 884–903, 1998.

[27] S. Green, M. B. Gavin, and L. Aiman-Smith, "Assessing a multidimensional measure of radical technological innovation," *IEEE Trans. Eng. Manag.*, vol. 42, no. 3, pp. 203–214, Aug. 1995.

[28] G. E. Greenley, "Market orientation and company performance: Empirical evidence from UK Companies," *Brit. J. Manage.*, vol. 6, pp. 1–13, 1995.

[29] A. Griffin and J. R. Hauser, "The voice of the customer," *Mark. Sci.*, vol. 12, no. 1, pp. 1–27, 1993.

[30] A. Griffin and A. L. Page, "An interim report on measuring product development success and failure," *J. Prod. Innov. Manage.*, vol. 10, no. 4, pp. 291–308, 1993.

[31] A. Grinstein, "The effect of market orientation and its components on innovation consequences: A meta-analysis," *J. Acad. Mark. Sci.*, vol. 36, pp. 166–173, 2008.

[32] J. F. Hair, R. E. Anderson, R. L. Tatham, and W. C. Black, *Multivariate Data Analysis*. Upper Saddle River, NJ: Prentice-Hall, 1998.

[33] J. K. Han, N. Kim, and R. K. Srivastava, "Market orientation and organizational performance: Is innovation a missing link?," *J. Mark.*, vol. 62, no. 4, pp. 30–45, 1998.

[34] D. H. Henard and D. M. Szymanski, "Why some new products are more successful than others," *J. Mark. Res.*, vol. 38, no. 3, pp. 362–375, 2001.

[35] G. Hult, M. Thomas, and D. J. Ketchen, "Does market orientation matter? A test of the relationship between positional advantage and performance," *Strateg. Manage. J.*, vol. 22, no. 9, pp. 899–906, 2001.
[36] C. Homburg and C. Pflesser, "A multiple-layer model of market-oriented organizational culture: Measurement issues and performance outcomes," *J. Mark. Res.*, vol. 37, pp. 449–462, Nov. 2000.
[37] S. Im and J. P. Workman, "Market orientation, creativity, and new product performance in high-technology firms," *J. Mark.*, vol. 68, pp. 114–132, Apr. 2004.
[38] S. Im, M. Hussain, and S. Sengupta, "Testing interaction effects of the dimensions of market orientation on marketing program creativity," *J. Bus. Res.*, vol. 61, pp. 859–867, 2008.
[39] B. J. Jaworski and A. K. Kohli, "Market orientation: Antecedents and consequences," *J. Mark.*, vol. 57, pp. 53–70, Jul. 1993.
[40] N. Jimenez-Jimenez and J. G. Cegarra-Navarro, "The performance effect of organizational learning and market orientation," *Ind. Mark. Manage.*, vol. 36, pp. 694–708, 2007.
[41] A. H. Kirca, S. Jayachandran, and W. O. Bearden, "Market orientation: A meta-analytic review and assessment of its antecedents and impact on performance," *J. Mark.*, vol. 69, no. 2, pp. 24–41, 2005.
[42] A. K. Kohli and B. J. Jaworski, "Market orientation: The construct, research propositions, and managerial implications," *J. Mark.*, vol. 54, pp. 1–18, Apr. 1990.
[43] N. Lado, A. Maydeu-Olivares, and J. Rivera, "Measuring market orientation in several populations: A structural equations model," *Eur. J. Mark.*, vol. 32, no. 1/2, pp. 23–39, 1998.
[44] F. Langerak, E. J. Hultink, and H. S. J. Robben, "'The impact of market orientation, product advantage and launch proficiency on new product performance and organizational performance'," *J. Prod. Innov. Manage.*, vol. 21, no. 2, pp. 79–94, 2004.
[45] R. Th. A. J. Leenders, L. M. L. van Engelen, and J. Kratzer, "Virtuality, communication, and new product team creativity: A social network perspective," *J. Eng. Technol. Manage.*, vol. 20, no. 1–2, pp. 69–92, 2003.
[46] D. Leonard and J. F. Rayport, "Sparking innovation through empathic design," *Harv. Bus. Rev.*, vol. 75, no. 6, pp. 102–113, 1997.
[47] B. Lukas and O. Ferrell, "The effect of market orientation on product innovation," *J. Acad. Mark. Sci.*, vol. 28, no. 2, pp. 239–247, 2000.
[48] D. W. Marquardt, "Generalized inverses, ridge regression, biased linear estimation, and nonlinear estimation," *Technometrics*, vol. 12, no. 3, pp. 591–612, 1970.
[49] E. F. McQuarrie, *Customer Visits: Building a Better Market Focus*. New York: M.E. Sharpe, 2008.
[50] A. Menon and P. R. Varadarajan, "A model of marketing knowledge use within firms," *J. Mark.*, vol. 56, no. 4, pp. 53–71, 1992.
[51] R. K. Moenaert, W. E. Souder, A. De Meyer, and D. Deschoolmeester, "R&D-marketing integration mechanisms, communication flows, and innovation success," *J. Prod. Innov. Manage.*, vol. 11, no. 1, pp. 31–45, 1994.
[52] C. Moorman, "Organizational market information processes: Cultural antecedents and new product outcomes," *J. Mark. Res.*, vol. 32, pp. 318–335, Aug. 1995.
[53] C. Moorman and A. S. Miner, "The impact of organizational memory on new product performance and creativity," *J. Mark. Res.*, vol. 34, no. 1, pp. 91–106, 1997.
[54] J. C. Narver and S. F. Slater, "The effect of a market orientation on business profitability," *J. Mark.*, vol. 54, pp. 20–35, Oct. 1990.
[55] J. C. Narver, S. F. Slater, and D. L. MacLachlan, "Responsive and proactive market orientation and new-product success," *J. Prod. Innov. Manage.*, vol. 25, no. 5, pp. 334–347, 2004.
[56] B. D. Ottum and W. L. Moore, "The role of market information in new product success/failure," *J. Prod. Innov. Manage.*, vol. 14, no. 4, pp. 258–273, 1997.
[57] P. M. Podsakoff, S. B. MacKenzie, J.-Y. Lee, and N. P. Podsakoff, "Common method biases in behavioral research: A critical review of the literature and recommended remedies," *J. Appl. Psychol.*, vol. 88, no. 5, pp. 879–903, 2003.
[58] R. W. Ruekert, "Developing a market orientation: An organizational strategy perspective," *Int. J. Res. Mark.*, vol. 9, pp. 225–245, 1992.
[59] J. M. Sinkula, W. E. Baker, and T. Noordewier, "A Framework for market-based organizational learning: Linking values, knowledge and behavior," *J. Acad. Mark. Sci.*, vol. 25, no. 4, pp. 305–318, 1997.
[60] M. Song, H. Bij, and M. Weggeman, "Determinants of the level of knowledge application: A knowledge-based and information-processing perspective," *J. Prod. Innov. Manage.*, vol. 22, pp. 430–444, 2005.
[61] M. Song and J. Noh, "Best new product development and management practices in the Korean high-tech industry," *Ind. Mark. Manage.*, vol. 35, no. 3, pp. 262–278, 2006.
[62] M. L. Tushman and D. A. Nadler, "Information processing as an integrating concept in organizational design," *Acad. Manage. Rev.*, vol. 3, no. 3, pp. 613–624, 1978.
[63] E. Veldhuizen, E. J. Hultink, and A. Griffin, "Modeling market information processing in new product development: An empirical analysis," *J. Eng. Technol. Manage.*, vol. 23, pp. 353–373, 2006.
[64] K. Weick, "Enactment processes in organizations," in *New Directions in Organizational Behavior*, B. M. Staw and G. R. Salancik, Eds. Chicago, IL: St. Clair, 1977, pp. 267–300.
[65] M. H. Zack, "Managing codified knowledge," *Sloan Manage. Rev.*, vol. 40, no. 4, pp. 45–58, 1999.
[66] D. Zahay and A. Griffin, "Customer learning processes, strategy selection, and performance in business-to-business service firms," *Decis. Sci.*, vol. 35, no. 2, pp. 169–204, 2004.
[67] D. Zahay, A. Griffin, and E. Fredericks, "Sources, uses and forms of data in the new product development process," *Ind. Mark. Manage.*, vol. 33, pp. 657–666, Oct. 2004.
[68] D. Zahay, J. Peltier, D. E. Schultz, and A. Griffin, "The role of transactional versus relational data in IMC programs: Bringing customer data together," *J. Advert. Res.*, vol. 44, no. 1, pp. 3–18, 2004.

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